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Fourth Semester, B.Tech Course work
Mid Semester Theory Examination, February, 2026

Course code: CAMTC13/COMTC13
Course title: Probability and Stochastic Processes

Time: 1 hour 30 minutes Maximum Marks. 25

Note: Attempt all questions in given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	In a group of engineering students, 60% know MATLAB, 50% know Python, and 72% know at least one of these two software tools. What percentage of students know Python among those who also know MATLAB?	2.5	CO1
1b	Show that the function $$f_X(x) = \begin{cases} k(1-x^2), & -1 \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$ is a probability density function. Find the value of k and obtain the distribution function of X.	2.5	CO1
2a	A discrete random variable X takes values $\{-2, 0, 2\}$ such that $P(X=-2) = P(X=2) = 2P(X=0)$. Find the mean of the random variable $Y = 3X + 4$.	2.5	CO1
2b	The probability density function of a random variable X is given by $$f_X(x) = \begin{cases} \frac{1}{a}, & 0 \leq x \leq a, \\ 0, & \text{otherwise.} \end{cases}$$ Find the variance of X.	2.5	CO1
3a	The lifetime (in hours) of a machine component is a continuous random variable X with p.d.f. $$f_X(x) = \begin{cases} \frac{200}{x^2}, & x \geq 200, \\ 0, & \text{otherwise.} \end{cases}$$ Find the probability that the component lasts less than 400 hours, given that it has already lasted 300 hours.	2.5	CO1
3b	A sample of 10 items is drawn at random from a box containing 60 items, of which 4 are defective. Find the probability that the sample contains at most one defective item.	2.5	CO2
4a	Derive the mean of a Poisson distribution with parameter λ.	2.5	CO2
4b	A random variable X is normally distributed with mean 50 and standard deviation 10. Find $P(40 \leq X \leq 70)$.	2.5	CO2
5a	State and apply Chebyshev's inequality to find an upper bound for $P(\lvert X-5 \rvert \geq 3)$, given that $E(X) = 5$ and $\mathrm{Var}(X) = 4$.	2.5	CO2
5b	Fit a Poisson distribution to the following data and obtain the expected frequencies. <table><tr><td>**x**</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>**f**</td><td>90</td><td>65</td><td>30</td><td>15</td></tr></table>	2.5	CO2